03 Alternating Currents (AC)

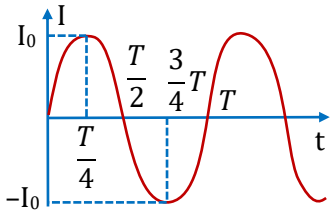
Alternating Current and Voltage

Voltage or current is said to be alternating if it is changed continuously in magnitude and periodically in direction with time. It can be represented by a sine curve or cosine curve

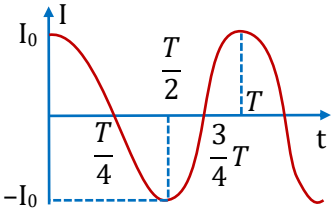
$I = I_0 \sin \omega t$ or $I = I_0 \cos \omega t$

where I = Instantaneous value of current at time t, I_0 = Amplitude or peak value

ω = Angular frequency $\omega = \frac{2\pi}{T} = 2\pi f$ T = time period f = frequency



I as a sine function of t



I as a cosine function of t

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Amplitude of AC:

The maximum value of current in either direction is called peak value or the amplitude of current. It is represented by I_0. Peak to peak value = $2I_0$.

Periodic Time:

The time taken by alternating current to complete one cycle of variation is called periodic time or time period of the current.

Frequency:

The number of cycles completed by an alternating current in one second is called the frequency of the current.

Unit : cycle/s ; (Hz)

In India : $f = 50$ Hz, supply voltage = 220 volt

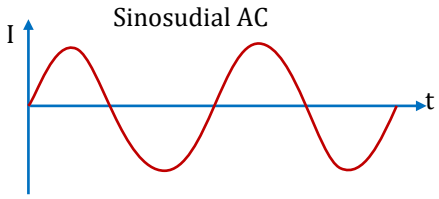
In USA : $f = 60$ Hz, supply voltage = 110 volt

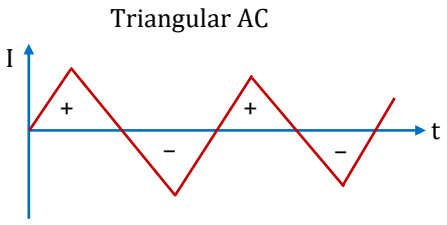
Condition required for current/ voltage to be Alternating:

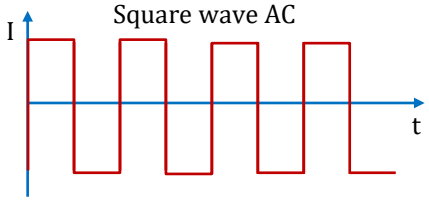
- Amplitude is constant.
- Alternate half cycle is positive and half negative.

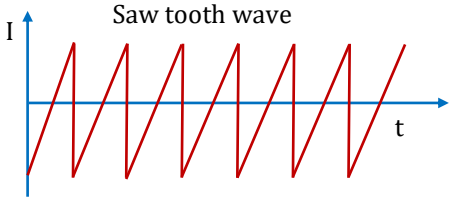
The alternating current continuously varies in magnitude and periodically reverses its direction.

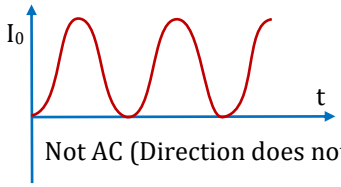
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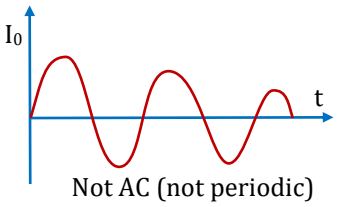


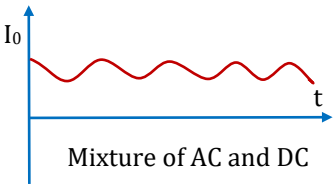












Average Value or Mean Value

The mean value of *AC* over any half cycle (either positive or negative) is that value of *DC* which would send same amount of charge through a circuit as is sent by the *AC* through same circuit in the same time.

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Average value of current for half cycle $< I > = \frac{\int_0^{T/2} Idt}{\int_0^{T/2} dt}$

Average value of $I = I_0 \sin \omega t$ over the positive half cycle :

$$I_{av} = \frac{\int_0^{\frac{T}{2}} I_0 \sin \omega t dt}{\int_0^{\frac{T}{2}} dt} = \frac{2I_0}{\omega T}\left[-\cos \omega t\right]_0^{\frac{T}{2}} = \frac{2I_0}{\pi} 1$$

$< \sin \theta > = < \sin 2\theta > = 0$
$< \cos \theta > = < \cos 2\theta > = 0$
$< \sin \theta \cos \theta > = 0$
$< \sin^2 \theta > = < \cos^2 \theta > = \frac{1}{2}$

- For symmetric *AC*, average value over full cycle = 0,

Average value of sinusoidal *AC*

Full cycle	(+ve) half cycle	(–ve) half cycle
0	$\frac{2I_0}{\pi}$	$-\frac{2I_0}{\pi}$

As the average value of AC over a complete cycle is zero, it is always defined over a half cycle which must be either positive or negative.

Maximum Value

- $I = a \sin\theta \Rightarrow I_{Max.} = a$
- $I = a \sin\theta + b \cos\theta \Rightarrow I_{Max.} = \sqrt{a^2 + b^2}$
- $I = a + b \sin\theta \Rightarrow I_{Max.} = a + b$ (if a and $b > 0$)
- $I = a \sin^2\theta \Rightarrow I_{Max.} = a$ ($a > 0$)

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Root Mean Square (rms) Value

It is value of *DC* which would produce same heat in given resistance in given time as is done by the alternating current when passed through the same resistance for the same time.

$$I_{rms} = \sqrt{\frac{\int_0^T I^2 dt}{\int_0^T dt}}$$ *rms* value = Virtual value = Apparent value

***rms* value of $I = I_0 \sin \omega t$:**

$$I_{rms} = \sqrt{\frac{\int_0^T (I_0 \sin \omega t)^2 dt}{\int_0^T dt}} = \sqrt{\frac{I_0^2}{T}\int_0^T \sin^2 \omega t \, dt}$$

$$= I_0 \sqrt{\frac{1}{T}\int_0^T \left[\frac{1-\cos 2\omega t}{2}\right] dt} = I_0 \sqrt{\frac{1}{T}\left[\frac{t}{2} - \frac{\sin 2\omega t}{2\times 2\omega}\right]_0^T} = \frac{I_0}{\sqrt{2}}$$

- If nothing is mentioned then values printed in *a.c.* circuit on electrical appliances, any given or unknown values, reading of *AC* meters are assumed to be *RMS*.

Current	Average	Peak	RMS	Angular frequency
$I_1 = I_0 \sin \omega t$	0	I_0	$\frac{I_0}{\sqrt{2}}$	ω
$I_2 = I_0 \sin \omega t \cos \omega t = \frac{I_0}{2}\sin 2\omega t$	0	$\frac{I_0}{2}$	$\frac{I_0}{2\sqrt{2}}$	2ω
$I_3 = I_0 \sin\omega t + I_0 \cos\omega t$	0	$\sqrt{2}\, I_0$	I_0	ω

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- For above varieties of current $rms = \frac{\text{Peak value}}{\sqrt{2}}$

Illustration 1

If $I = 2\sqrt{t}$ ampere then calculate average and rms values over t = 2 to 4*s*.

Solution:

$$\langle I \rangle = \frac{\int_2^4 2\sqrt{t}.dt}{\int_2^4 dt} = \frac{4}{3}\frac{(t^{\frac{3}{2}})_2^4}{(t)_2^4} = \frac{2}{3}\left[8 - 2\sqrt{2}\right] \quad \text{and} \quad I_{rms} = \sqrt{\frac{\int_2^4 (2\sqrt{t})^2 dt}{\int_2^4 dt}} = \sqrt{\frac{\int_2^4 4t dt}{2}} = \sqrt{2\left[\frac{t^2}{2}\right]_2^4} = 2\sqrt{3}$$

Illustration 2

Find the time required for 50Hz alternating current to change its value from zero to rms value.

Solution:

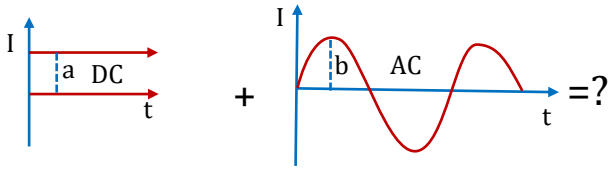
$\because \quad I = I_0 \sin \omega t \quad \therefore \quad \frac{I_0}{\sqrt{2}} = I_0 \sin \omega t \;\Rightarrow\; \sin \omega t = \frac{1}{\sqrt{2}} \Rightarrow \omega t = \frac{\pi}{4}$

$\Rightarrow \quad \left(\frac{2\pi}{T}\right) t = \frac{\pi}{4} \quad \Rightarrow t = \frac{T}{8} = \frac{1}{8\times 50} = 2.5 \; ms$

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Illustration 3

If a direct current of value a ampere is superimposed on an alternating current $I = b \sin \omega t$ flowing through a wire, what is the effective value of the resulting current in the circuit?



Solution:

As current at any instant in the circuit will be,

$$I = I_{DC} + I_{AC} = a + b \sin\omega t$$

$$\therefore \quad I_{eff} = \sqrt{\frac{1}{T}\int_0^T I^2 dt} = \sqrt{\frac{1}{T}\int_0^T (a + b\sin\omega t)^2 dt} = \sqrt{\frac{1}{T}\int_0^T (a^2 + 2ab\sin\omega t + b^2\sin^2\omega t) dt}$$

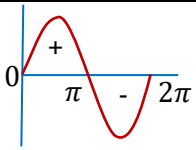
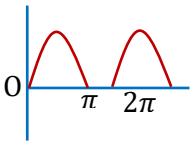
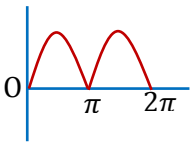
But as $\frac{1}{T}\int_0^T \sin\omega t dt = 0$

and $\frac{1}{T}\int_0^T \sin^2\omega t dt = \frac{1}{2}$

$$\therefore \quad I_{eff} = \sqrt{a^2 + \frac{1}{2}b^2}$$

Some Important wave forms and their RMS and Average Value:

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Nature of wave form	Wave–form	RMS Value	Average or mean Value
Sinusoidal		$\frac{I_0}{\sqrt{2}} = 0.707\ I_0$	$\frac{2I_0}{\pi} = 0.637\ I_0$ (Half Cycle) Zero (Full Cycle)
Half wave rectified		$\frac{I_0}{2} = 0.5\ I_0$	$\frac{I_0}{\pi} = 0.318\ I_0$
Full wave rectified		$\frac{I_0}{\sqrt{2}} = 0.707\ I_0$	$\frac{2I_0}{\pi} = 0.637\ I_0$

Phase and phase difference

(a) Phase :

$I = I_0 \sin(\omega t \pm \phi)$

Initial phase = ϕ (it does not change with time)

Instantaneous phase = $\omega t \pm \phi$ (it changes with time)

- Phase decides both value and sign. ◎ **Unit:** radian

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(b) Phase difference :

Voltage $V = V_0 \sin(\omega t + \phi_1)$ Current $I = I_0 \sin(\omega t + \phi_2)$

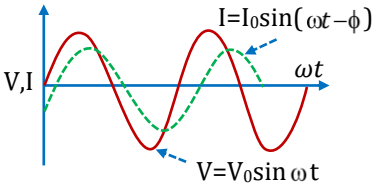
- Phase difference of I w.r.t. V $\phi = \phi_2 - \phi_1$
- Phase difference of V w.r.t. I $\phi = \phi_1 - \phi_2$

Lagging and leading Concept:

(a) V leads I or I lags $V \rightarrow$ It means, V reach maximum before I

Let if $V = V_0 \sin \omega t$ then $I = I_0 \sin(\omega t - \phi)$

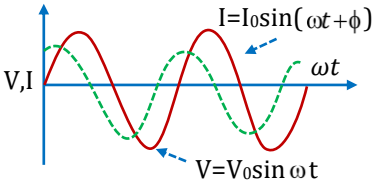
and if $V = V_0 \sin(\omega t + \phi)$ then $I = I_0 \sin \omega t$



(b) V lags I or I leads $V \rightarrow$ It means V reach maximum after I

Let if $V = V_0 \sin \omega t$ then $I = I_0 \sin(\omega t + \phi)$

and if $V = V_0 \sin(\omega t - \phi)$ then $I = I_0 \sin \omega t$



Phasor and Phasor diagram

A diagram representing alternating current and voltage (of same frequency) as vectors (phasor) with the phase angle between them is called phasor diagram.

Let $V = V_0 \sin \omega t$ and $I = I_0 \sin(\omega t + \phi)$

In figure (a) two arrows represents phasors. The length of phasors represents the maximum value of quantity. The projection of a phasor on y-axis represents the instantaneous value of quantity

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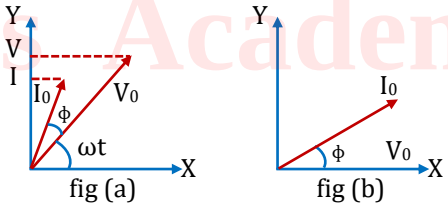


Illustration 4

The Equation of current in AC circuit is $I = 4\sin\left[100\pi t + \frac{\pi}{3}\right]$ A. Calculate.

(i) RMS Value (ii) Peak Value (iii) Frequency (iv) Initial phase (v) Current at $t = 0$

Solution:

(i) $I_{rms} = \frac{I_0}{\sqrt{2}} = \frac{4}{\sqrt{2}} = 2\sqrt{2}A$

(ii) Peak value $I_0 = 4A$

(iii) $\because$ $\omega = 100\,\pi\, rad/s$

$\therefore$ Frequency $f = \frac{100\pi}{2\pi} = 50\ Hz$

(iv) Initial phase $= \frac{\pi}{3}$

(v) At $t = 0$, $I = 4\sin\left[100\pi \times 0 + \frac{\pi}{3}\right] = 4 \times \frac{\sqrt{3}}{2} = 2\sqrt{3}\ A$

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Illustration 5

If $I = I_0 \sin \omega t$, $E = E_0 \cos\left[\omega t + \frac{\pi}{3}\right]$. Calculate phase difference between E and I.

Solution:

$I = I_0 \sin \omega t$ and $E = E_0 \sin\left[\frac{\pi}{2} + \omega t + \frac{\pi}{3}\right]$

$\therefore$ Phase difference = $\frac{\pi}{2} + \frac{\pi}{3} = \frac{5\pi}{6}$

Different types of *AC* circuits:

In order to study the behaviour of *A.C.* circuits we classify them into two categories :

(a) Simple circuits containing only one basic element i.e. resistor (R) or inductor (L) or capacitor (C) only.

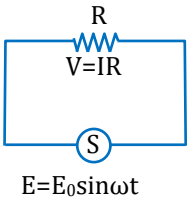
(b) Complicated circuit containing any two of the three circuit elements R, L and C or all of three elements.

AC Circuit Containing Pure Resistance:

Let at any instant the current in the circuit = I.

Potential difference across the resistance = IR.

with the help of Kirchoff's circuital law $E - IR = 0$



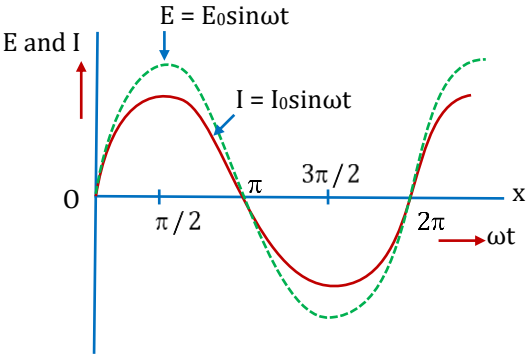
$\Rightarrow \quad E_0 \sin \omega t = IR$

$\Rightarrow \quad I = \frac{E_0}{R} \sin \omega t = I_0 \sin \omega t$

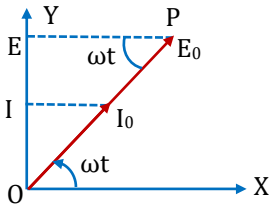
($I_0 = \frac{E_0}{R}$ = peak or maximum value of current)

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Alternating current developed in a pure resistance is also of sinusoidal nature. In an *a.c.* circuits containing pure resistance, the voltage and current are in the same phase. The vector or phasor diagram which represents the phase relationship between alternating current and alternating *e.m.f.* as shown in figure.



In the *a.c.* circuit having R only, as current and voltage are in the same phase, hence in fig. both phasors E_0 and I_0 are in the same direction, making an angle ωt with OX. Their projections on Y-axis represent the instantaneous values of alternating current and voltage.



i.e. $I = I_0 \sin \omega t$ and $E = E_0 \sin \omega t$

Since $I_0 = \frac{E_0}{R}$, Hence $\frac{I_0}{\sqrt{2}} = \frac{E_0}{R\sqrt{2}}$

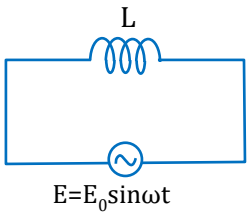
$\Rightarrow \quad I_{rms} = \frac{E_{rms}}{R}$

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AC circuit containing pure inductance:

A circuit containing a pure inductance L (having zero ohmic resistance) connected with a source of alternating emf.

Let the alternating e.m.f., $E = E_0 \sin \omega t$



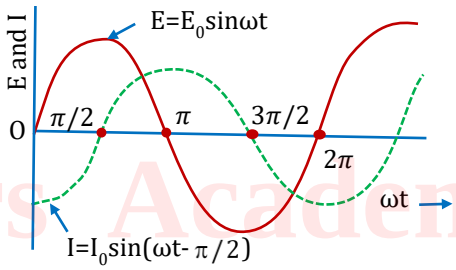
When *a.c.* flows through the circuit, emf induced across inductance, $E = -L\frac{dI}{dt}$

Negative sign indicates that induced *emf* acts in opposite direction to that of applied *emf*.

Because there is no other circuit element present in the circuit other than inductance so with the help of Kirchhoff's circuital law $E + \left(-L\frac{dI}{dt}\right) = 0 \Rightarrow E = L\frac{dI}{dt}$ so we get $I = \frac{E_0}{\omega L}\sin\left(\omega t - \frac{\pi}{2}\right)$

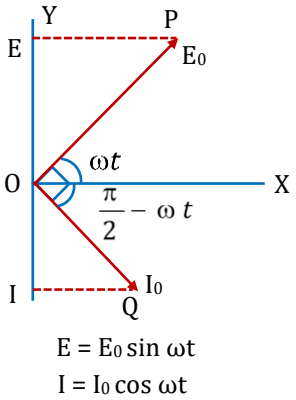
Maximum current, $I_0 = \frac{E_0}{\omega L} \times 1 = \frac{E_0}{\omega L}$, hence, $I = I_0 \sin\left(\omega t - \frac{\pi}{2}\right)$

In a pure inductive circuit current always lags behind the emf by $\frac{\pi}{2}$.



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Or alternating *emf* leads the *a. c.* by a phase angle of $\frac{\pi}{2}$.



$E = E_0 \sin \omega t$

$I = I_0 \cos \omega t$

Expression $I_0 = \frac{E_0}{\omega L}$ resembles the expression $\frac{E}{I} = R$.

This non-resistive opposition to the flow of *A.C.* in a circuit is called the inductive reactance (X_L) of the circuit.

$X_L = \omega L = 2\pi f L$ where f = frequency of *A.C.*

Unit of X_L : ohm

(ωL) = Unit of L × Unit of ω = henry × $\sec^{-1}$

$$= \frac{\text{Volt}}{\text{Ampere/sec}} \times \sec^{-1} = \frac{\text{Volt}}{\text{Ampere}} = \text{ohm}$$

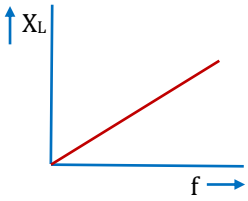
Inductive reactance $X_L \propto f$

Higher the frequency of *A.C.* higher is the inductive reactance offered by an inductor in an *A.C.* circuit.

For *d.c.* circuit, $f = 0$

$\therefore \quad X_L = \omega L = 2\pi f L = 0$

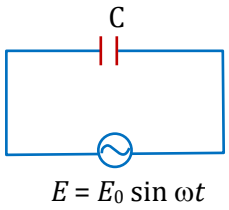
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Hence, inductor offers no opposition to the flow of *d.c.* whereas a resistive path to *a.c.*

AC circuit containing pure capacitor

A circuit containing an ideal capacitor of capacitance C connected with a source of alternating *emf* as shown in fig. The alternating *e.m.f.* in the circuit $E = E_0 \sin \omega t$



When alternating *e.m.f.* is applied across the capacitor a similarly varying alternating current flows in the circuit.

The two plates of the capacitor become alternately positively and negatively charged and the magnitude of the charge on the plates of the capacitor varies sinusoidally with time. Also, the electric field between the plates of the capacitor varies sinusoidally with time. Let, at any instant charge on the capacitor = q.

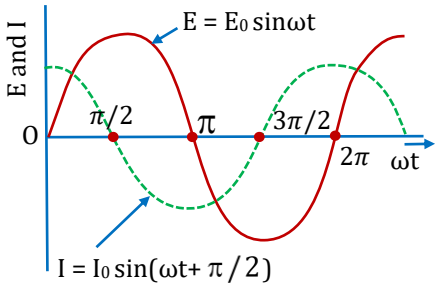
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Instantaneous potential difference across the capacitor,

$$E = \frac{q}{C}$$

$$\Rightarrow \quad q = C\,E$$

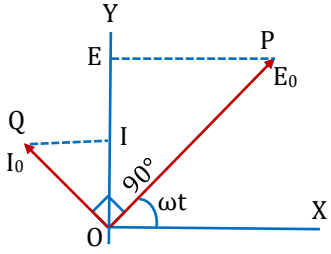
$$\Rightarrow \quad q = CE_0 \sin \omega t$$



The instantaneous value of current

$$I = \frac{dq}{dt} = \frac{d}{dt}\left(CE_0 \sin \omega t\right) = CE_0 \omega \cos \omega t$$

$$\Rightarrow \quad I = \frac{E_0}{(1/\omega C)} \sin\left(\omega t + \frac{\pi}{2}\right) = I_0 \sin\left(\omega t + \frac{\pi}{2}\right)$$



where $I_0 = \omega C V_0$

In a pure capacitive circuit, the current always leads the *e.m.f.* by a phase angle of $\pi/2$. The alternating *emf* lags behinds the alternating current by a phase angle of $\pi/2$.

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Important Points

$\frac{E}{I}$ is the resistance R when both E and I are in phase, in present case they differ in phase by $\frac{\pi}{2}$, hence $\frac{1}{\omega C}$ is not the resistance of the capacitor, the capacitor offer opposition to the flow of *A.C.* This non-resistive opposition to the flow of *A.C.* in a pure capacitive circuit is known as capacitive reactance $X_C = \frac{1}{\omega C} = \frac{1}{2\pi f C}$.

Unit of X_C : ohm

Capacitive reactance X_C is inversely proportional to frequency of *A.C.* X_C decreases as the frequency increases.

This is because with an increase in frequency, the capacitor charges and discharges rapidly following the flow of current.

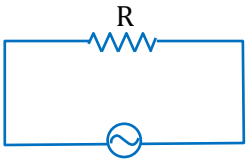
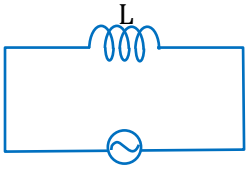
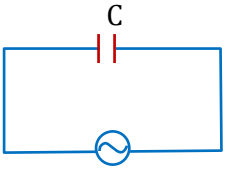
For *d.c.* circuit $f = 0$

$\therefore \quad X_C = \frac{1}{2\pi f C} = \infty$, but has a very small value for *a.c.*

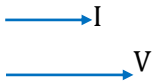
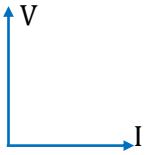
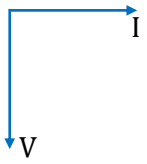
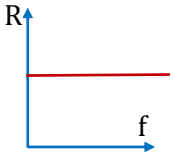
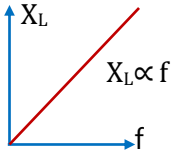
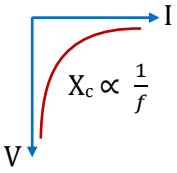
This shows that capacitor blocks the flow of *d.c.* but provides an easy path for *a.c.*

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Individual Components (R or L or C)

Term	R	L	C
Circuit			
Supply Voltage	$V = V_0 \sin \omega t$	$V = V_0 \sin \omega t$	$V = V_0 \sin \omega t$
Current	$I = I_0 \sin \omega t$	$I = I_0 \sin (\omega t - \frac{\pi}{2})$	$I = I_0 \sin (\omega t + \frac{\pi}{2})$
Peak Current	$I_0 = \frac{V_0}{R}$	$I_0 = \frac{V_0}{\omega L}$	$I_0 = \frac{V_0}{1/\omega C} = V_0 \omega C$
Impedance (Ω)	$\frac{V_0}{I_0} = R$	$\frac{V_0}{I_0} = \omega L = X_L$	$\frac{V_0}{I_0} = \frac{1}{\omega C} = X_C$
$Z = \frac{V_0}{I_0} = \frac{V_{rms}}{I_{rms}}$	R = Resistance	X_L=Inductive reactance	X_C=Capacitive reactance
Phase difference	zero (in same phase)	$+\frac{\pi}{2}$ (V leads I)	$-\frac{\pi}{2}$ (V lags I)

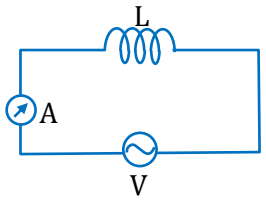
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Term	R	L	C
Phasor diagram			
Variation of *Z* with *f*	 *R* does not depend on *f*		
***G*,S_L,S_C (mho, seiman)**	$G=1/R$ = conductance	$S_L = 1/X_L$ Inductive susceptance	$S_C = 1/X_C$ Capacitive susceptance
Behaviour of device in *D.C.* and *A.C*	Same in *A C* and *D C*	*L* passes *DC* easily (because $X_L = 0$) while gives a high impedance for the *A.C.* of high frequency ($X_L \propto f$)	*C* - blocks *DC* (because $X_C = \infty$) while provides an easy path for the *A.C.* of high frequency $\left[X_C \propto \frac{1}{f}\right]$
Ohm's law	$V_R = IR$	$V_L = IX_L$	$V_C = IX_C$

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Illustration 6

In given circuit applied voltage $V = 50 \sin 100\pi t$ volt and ammeter reading is $2A$ then calculate value of L.



Solution:

$$V_{rms} = I_{rms} X_L$$

$\because$ Reading of ammeter = I_{rms}

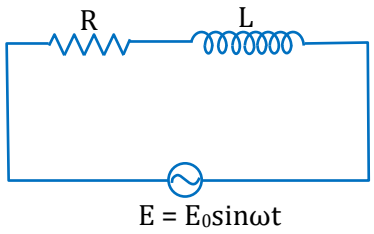
$$X_L = \frac{V_{rms}}{I_{rms}} = \frac{V_0}{\sqrt{2} I_{rms}} = \frac{50\sqrt{2}}{\sqrt{2} \times 2} = 25\ \Omega$$

$$\Rightarrow \quad L = \frac{X_L}{\omega} = \frac{25}{100\pi} = \frac{1}{4\pi}\ H$$

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Resistance and Inductance in Series (R-L Circuit)

A circuit containing a series combination of a resistance R and an inductance L, connected with a source of alternating *e.m.f.* E as shown in figure.



Phasor Diagram for L-R Circuit

Let in a L-R series circuit, applied alternating emf is $E = E_0 \sin \omega t$. As R and L are joined in series, hence current flowing through both will be same at each instant. Let I be the current in the circuit at any instant and V_L and V_R the potential differences across L and R respectively at that instant.

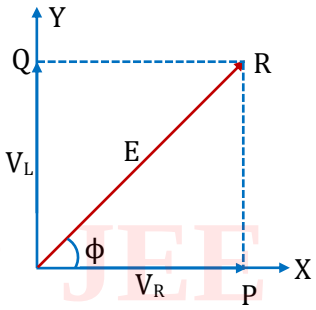
Then $V_L = IX_L$ and $V_R = IR$

Now, V_R is in phase with the current while V_L leads the current by .

So V_R and V_L are mutually perpendicular (Note : $E \neq V_R + V_L$)

The vector OP represents V_R (which is in phase with I),

while OQ represents V_L (which leads I by 90°).

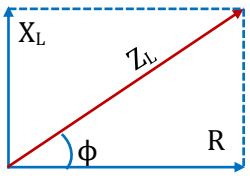


The resultant of V_R and V_L = the magnitude of vector OR $E = \sqrt{V_R^2 + V_L^2}$

Thus $E^2 = V_R^2 + V_L^2 = I^2 (R^2 + X_L^2)$

$$\Rightarrow \quad I = \frac{E}{\sqrt{R^2 + X_L^2}}$$

The phasor diagram shown in fig. also shows that in L-R circuit the applied emf E leads the current I or conversely the current I lags behind the *e.m.f.* E. by a phase angle ϕ $\tan\phi = \frac{V_L}{V_R} = \frac{IX_L}{IR} = \frac{X_L}{R} = \frac{\omega L}{R}$



$$\Rightarrow \quad \phi = \tan^{-1}\left(\frac{\omega L}{R}\right)$$

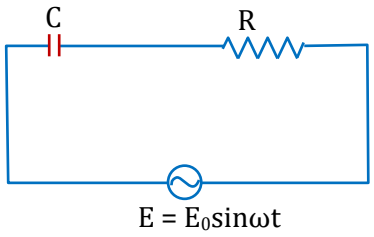
Inductive Impedance Z_L:

In L-R circuit the maximum value of current $I_0 = \frac{E_0}{\sqrt{R^2 + \omega^2 L^2}}$. Here $\sqrt{R^2 + \omega^2 L^2}$ represents the effective opposition offered by L-R circuit to the flow of *a.c.* through it. It is known as impedance of L-R circuit and is represented by Z_L. $Z_L = \sqrt{R^2 + \omega^2 L^2} = \sqrt{R^2 + (2\pi f L)^2}$. The reciprocal of impedance is called admittance $Y_L = \frac{1}{Z_L} = \frac{1}{\sqrt{R^2 + \omega^2 L^2}}$.

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Resistance and Capacitor in Series (*R-C* Circuit)

A circuit containing a series combination of a resistance R and a capacitor C, connected with a source of *e.m.f.* of peak value E_0 as shown in fig.

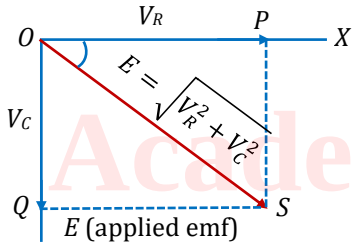


Phasor Diagram for R-C Circuit

Current through both the resistance and capacitor will be same at every instant and the instantaneous potential differences across C and R are

$V_C = IX_C$ and $V_R = IR$

where X_C = capacitive reactance and I = instantaneous current.



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Now, V_R is in phase with I, while V_C lags behind I by 90°.

The phasor diagram is shown in fig.

The vector OP represents V_R (which is in phase with I) and the vector OQ represents V_C (which lags behind I by $\frac{\pi}{2}$).

The vector OS represents the resultant of V_R and V_C = the applied *e.m.f.* E.

Hence $V_R^2 + V_C^2 = E^2$ $\Rightarrow E = \sqrt{V_R^2 + V_C^2}$

$\Rightarrow$ $E^2 = I^2 (R^2 + X_C^2)$ $\Rightarrow I = \dfrac{E}{\sqrt{R^2 + X_C^2}}$

The term $\sqrt{(R^2 + X_C^2)}$ represents the effective resistance of the *R-C* circuit and called the capacitive impedance Z_C of the circuit.

Hence, in *C-R* circuit, $Z_C = \sqrt{R^2 + X_C^2} = \sqrt{R^2 + \left(\dfrac{1}{\omega C}\right)^2}$

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Capacitive Impedance Z_C :

In *R-C* circuit the term $\sqrt{R^2+X_C^2}$ effective opposition offered by *R-C* circuit to the flow of *a.c.* through it.

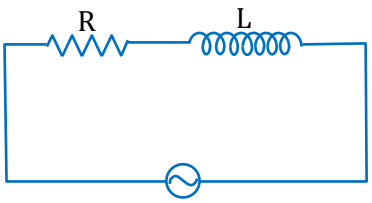
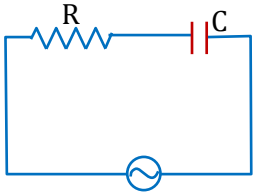
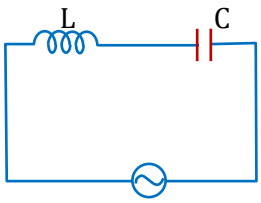
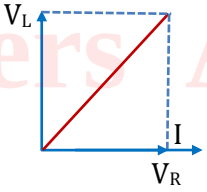
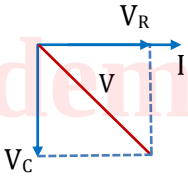
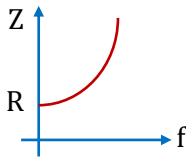
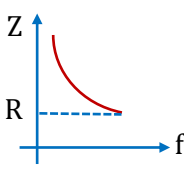
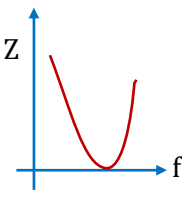
It is known as impedance of *R-C* circuit and is represented by Z_C.

The phasor diagram also shows that in *R-C* circuit the applied *e.m.f.* lags behind the current *I* (or the current *I* leads the *emf E*) by a phase angle ϕ given by

$$\tan\phi=\frac{V_C}{V_R}=\frac{X_C}{R}=\frac{1/\omega C}{R}=\frac{1}{\omega CR},\quad \tan\phi=\frac{X_C}{R}=\frac{1}{\omega CR}$$

$$\Rightarrow \qquad \phi=\tan^{-1}\left(\frac{1}{\omega CR}\right)$$

Combination of Components (*R-L* or *R-C* or *L-C*)

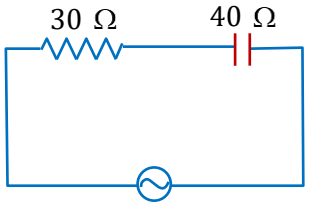
Term	*R-L*	*R-C*	*L-C*
Circuit	 *I* is same in R & L	 *I* is same in R & C	 *I* is same in L & C
Phasor diagram	 $V^2=V_R^2+V_L^2$	 $V^2=V_R^2+V_C^2$	 $V=V_L-V_C\ (V_L>V_C)$ $V=V_C-V_L\ (V_C>V_L)$
Phase difference between *V* and *I*	*V* leads *I* (ϕ = 0 to $\frac{\pi}{2}$)	*V* lags *I* ($\phi=-\frac{\pi}{2}$ to 0)	*V* lags *I* ($\phi=-\frac{\pi}{2}$, if $X_C>X_L$) *V* leads *I* ($\phi=\frac{\pi}{2}$, if $X_L>X_C$)
Variation of *Z*	$Z=\sqrt{R^2+(X_L)^2}$ as $f\uparrow, Z\uparrow$	$Z=\sqrt{R^2+(X_C)^2}$ as $f\uparrow, Z\downarrow$	$Z=\lvert X_L-X_C\rvert$ as $f\uparrow$, *Z* first $\downarrow$ then $\uparrow$
with *f*			
At very low *f*	$Z\simeq R\ (X_L\to 0)$	$Z\simeq X_C$	$Z\simeq X_C$
At very high *f*	$Z\simeq X_L$	$Z\simeq R\ (X_C\to 0)$	$Z\simeq X_L$

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Illustration 7

Calculate the impedance of the circuit shown in the figure.



Solution:

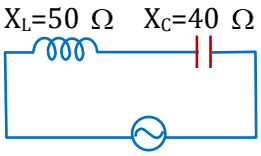
$$Z = \sqrt{R^2 + (X_c)^2}$$

$$= \sqrt{(30)^2 + (40)^2}$$

$$= \sqrt{2500} = 50\ \Omega$$

Illustration 8

If $X_L = 50\ \Omega$ and $X_C = 40\ \Omega$ Calculate effective value of current in given circuit.



Solution:

$$Z = X_L - X_C = 10\ \Omega$$

$$I_0 = \frac{V_0}{Z} = \frac{40}{10} = 4A$$

$$\Rightarrow \quad I_{rms} = \frac{4}{\sqrt{2}} = 2\sqrt{2}\ A$$

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Illustration 9

When an alternating voltage of $220V$ is applied across a device X, a current of $0.5\ A$ flows through the circuit and is in phase with the applied voltage. When the same voltage is applied across another device Y, the same current again flows through the circuit but it leads the applied voltage by $\pi/2$ radians.

(a) Name the devices X and Y.

(b) Calculate the current flowing in the circuit when same voltage is applied across the series combination of X and Y.

Solution:

(a) X is resistor and Y is a capacitor

(b) Since the current in the two devices is the same ($0.5A$ at 220 *volt*)

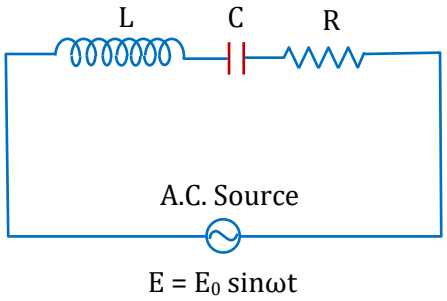
When R and C are in series across the same voltage then

$$R = X_C = \frac{220}{0.5} = 440\ \Omega \quad \Rightarrow I_{rms} = \frac{V_{rms}}{\sqrt{R^2 + X_C^2}} = \frac{220}{\sqrt{(440)^2 + (440)^2}} = \frac{220}{440\sqrt{2}} = 0.35A$$

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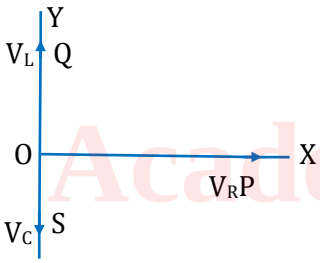
Inductance, Capacitance and Resistance in series (*l-C-R* series circuit)

A circuit containing a series combination of an resistance R, a coil of inductance L and a capacitor of capacitance C, connected with a source of alternating *e.m.f.* of peak value of E_0, as shown in fig.



Phasor Diagram for Series L-C-R Circuit

Let in series *LCR* circuit applied alternating *emf* is $E = E_0 \sin \omega t$. As L, C and R are joined in series, therefore, current at any instant through the three elements has the same amplitude and phase.



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However, voltage across each element bears a different phase relationship with the current.

Let at any instant of time t the current in the circuit is I.

Let at this time t the potential differences across L, C, and R

$V_L = IX_L$, $V_C = IX_C$ and $V_R = IR$

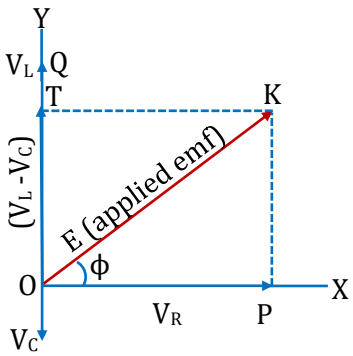
Now, V_R is in phase with current I but V_L leads I by 90°.

While V_C lags behind I by 90°.

The vector OP represents V_R (which is in phase with I) the vector OQ represent V_L (which leads I by 90°) and the vector OS represents V_C (which lags behind I by 90°) V_L and V_C are opposite to each other.

If $V_L > V_C$ (as shown in figure) then their resultant will be $(V_L - V_C)$ which is represented by OT.

Finally, the vector OK represents the resultant of V_R and $(V_L - V_C)$, that is, the resultant of all the three = applied *e.m.f.*



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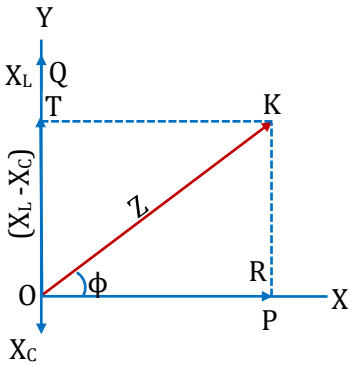
Thus $E = \sqrt{V_R^2 + (V_L - V_C)^2}$

$= I\sqrt{R^2 + (X_L - X_C)^2}$

$\Rightarrow \quad I = \dfrac{E}{\sqrt{R^2 + (X_L - X_C)^2}}$

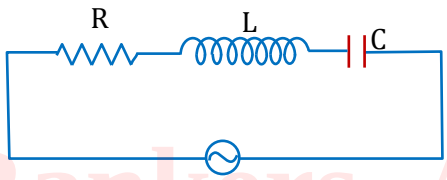
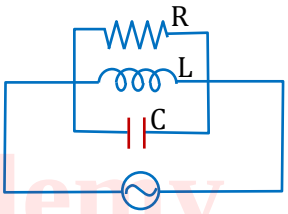
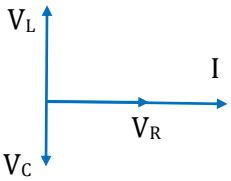
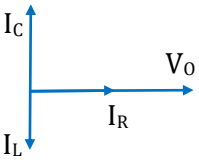
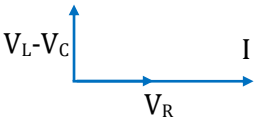
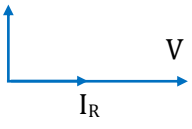
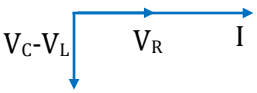
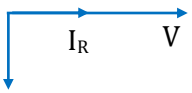
Impedance $Z = \sqrt{R^2 + (X_L - X_C)^2}$

$= \sqrt{R^2 + \left(\omega L - \dfrac{1}{\omega C}\right)^2}$



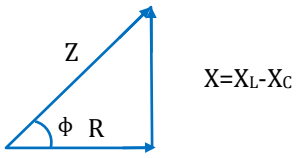
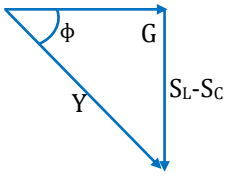
The phasor diagram also shown that in *LCR* circuit the applied *e.m.f.* leads the current *I* by a phase angle ϕ, tan ϕ = $\dfrac{X_L - X_C}{R}$

Series LCR and Parallel LCR Combination

	Series *L-C-R* Circuit		Parallel *L-C-R* Circuit
1.	**Circuit diagram** 		*V* same for *R*, *L* & *C*  V same for R,L and C
2.	**Phasor diagram** 		
(i)	If $V_L > V_C$ then 	**(i)**	if $I_C > I_L$ then 
(ii)	If $V_C > V_L$ then 	**(ii)**	if $I_L > I_C$ then 
(iii)	$V = \sqrt{V_R^2 + (V_L - V_C)^2}$ Impedance $Z = \sqrt{R^2 + (X_L - X_C)^2}$	**(iii)**	$I = \sqrt{I_R^2 + (I_L - I_C)^2}$ Admittance $Y = \sqrt{G^2 + (S_L - S_C)^2}$

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	Series *L-C-R* Circuit		Parallel *L-C-R* Circuit
	$\tan\phi = \frac{X_L - X_C}{R} = \frac{V_L - V_C}{V_R}$		$\tan\phi = \frac{S_L - S_C}{G} = \frac{I_L - I_C}{I_R}$
(iv)	Impedance triangle 	**(iv)**	Admittance triangle 

Golden Key Points

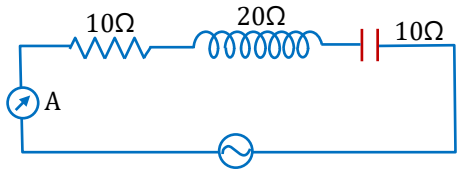
	Series		Parallel
(a)	if $X_L > X_C$ then V leads I, ϕ (positive) circuit nature inductive	**(a)**	if $S_L > S_C$ ($X_L < X_C$) then V leads I,ϕ (positive) circuit nature inductive
(b)	if $X_C > X_L$ then V lags I,ϕ (negative) circuit nature capacitive	**(b)**	if $S_C > S_L$ ($X_C < X_L$) then V lags I, ϕ (negative) circuit nature capacitive

- In A.C. circuit voltage for L or C may be greater than source voltage or current but it happens only when circuit contains L and C both and on R it is never greater than source voltage or current.
- In parallel A.C. circuit phase difference between I_L and I_C is π.

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Illustration 10

Find out reading of *AC* ammeter and also calculate the potential difference across, resistance and capacitor.



E = 100 sin100 π t volt

Solution:

$Z = \sqrt{R^2 + (X_L - X_C)^2} = 10\sqrt{2}\,\Omega \Rightarrow I_0 = \frac{V_0}{Z} = \frac{100}{10\sqrt{2}} = \frac{10}{\sqrt{2}} A$

$\because$ Ammeter reads *RMS* value, so its reading = $\frac{10}{\sqrt{2}\sqrt{2}} = 5A$

So, $V_R = 5 \times 10 = 50\ V$ and $V_C = 5 \times 10 = 50\ V$

Resonance

A circuit is said to be resonant when the natural frequency of circuit is equal to frequency of the applied voltage.

For resonance both L and C must be present in circuit.

There are two types of resonance : (i) Series Resonance (ii) Parallel Resonance

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Series Resonance

(a) At Resonance

(i) $X_L = X_C$ (ii) $V_L = V_C$

(iii) $f = 0$ (V and I in same phase) (iv) $Z_{min} = R$ (impedance minimum)

(v) $I_{max} = \frac{V}{R}$ (current maximum)

(b) Resonance frequency

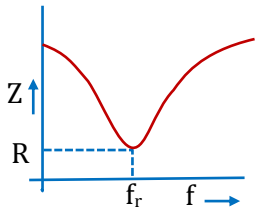
$$\because X_L = X_C \quad \Rightarrow \omega_r L = \frac{1}{\omega_r C} \Rightarrow \omega_r^2 = \frac{1}{LC} \Rightarrow \omega_r = \frac{1}{\sqrt{LC}} \Rightarrow f_r = \frac{1}{2\pi\sqrt{LC}}$$

(c) Variation of Z with f

(i) If $f < f_r$ then $X_L < X_C$ circuit nature is capacitive, ϕ (negative)

(ii) At $f = f_r$ then $X_L = X_C$ circuit nature is Resistive, ϕ = zero

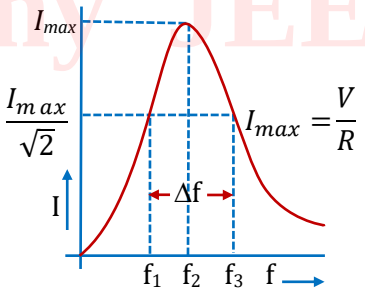
(iii) If $f > f_r$ then $X_L > X_C$ circuit nature is inductive, ϕ (positive)



Variation of *I* with *f* as f increase, Z first decreases then increases.

(d) As *f* increase, *I* first increase then decreases

At resonance impedance of the series resonant circuit is minimum so it is called 'acceptor circuit' as it most readily accepts that current out of many currents whose frequency is equal to its natural frequency. In radio or TV tuning we receive the desired station by making the frequency of the circuit equal to that of the desired station.



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Half power frequencies

The frequencies at which, power become half of its maximum value called half power frequencies

Band width = $\Delta f = f_2 - f_1$

Quality factor Q: Q-factor of AC circuit basically gives an idea about stored energy & lost energy.

$$Q = 2\pi \frac{\text{maximum energy stored per cycle}}{\text{maximum energy loss per cycle}}$$

(i) It represents the sharpness of resonance.

(ii) It is unit less and dimension less quantity.

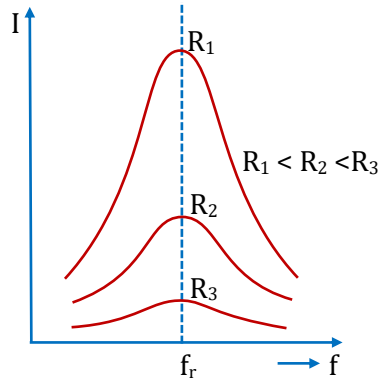
(iii) $$Q = \frac{(X_L)_r}{R} = \frac{(X_C)_r}{R} = \frac{2\pi f_r L}{R} = \frac{1}{R}\sqrt{\frac{L}{C}} = \frac{f_r}{\Delta f} = \frac{f_r}{\text{band width}}$$

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Sharpness

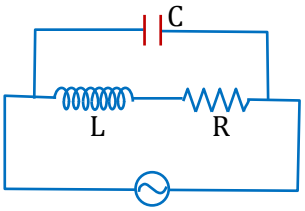
Sharpness $\propto$ Quality factor

R decrease $\Rightarrow Q$ increases $\Rightarrow$ Sharpness increases



Parallel Resonance

(a) At resonance



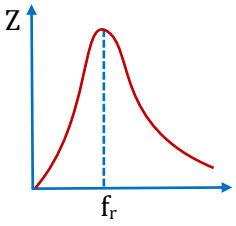
(i) $S_L = S_C$ (ii) $I_L = I_C$

(iii) $\phi = 0$ (iv) $Z_{max} = R$ (impedance maximum)

(v) $I_{min} = \frac{V}{R}$ (current minimum)

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(b) Resonant frequency $f_r = \frac{1}{2\pi\sqrt{LC}}$

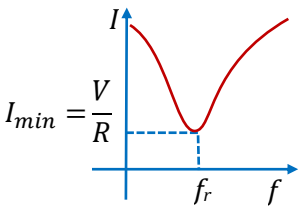


(c) Variation of Z with f as f increases, z first increases then decreases

◎ If $f < f_r$ then $S_L > S_C$, f (positive), circuit nature is inductive

◎ If $f > f_r$ then $S_C > S_L$, f (negative), circuit nature is capacitive.

(d) Variation of I with f as f increases, I first decreases then increases



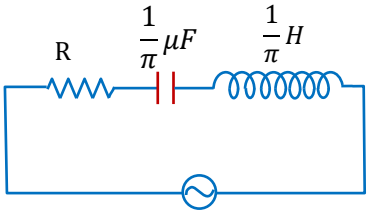
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Note:

For this circuit $f_r = \frac{1}{2\pi}\sqrt{\frac{1}{LC} - \frac{R^2}{L^2}} \Rightarrow Z_{max} = \frac{L}{RC}$ $\Rightarrow$ For resonance $\frac{1}{LC} > \frac{R^2}{L^2}$

Illustration 11

For what frequency the voltage across the resistance R will be maximum.



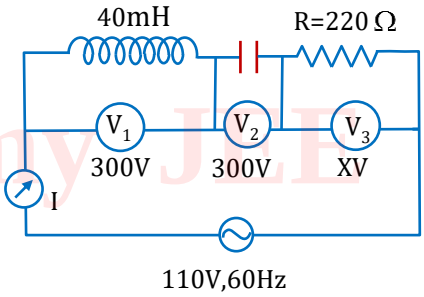
Solution:

It happens at resonance

$$f = \frac{1}{2\pi\sqrt{LC}} = \frac{1}{2\pi\sqrt{\frac{1}{\pi}\times 10^{-6}\times\frac{1}{\pi}}} = 500$$

Illustration 12

A capacitor, a resistor and a 40 mH inductor are connected in series to an AC source of frequency $60Hz$, calculate the capacitance of the capacitor, if the current is in phase with the voltage. Also calculate the value of X and I.



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Solution:

At resonance,

$$\omega L = \frac{1}{\omega C}, C = \frac{1}{\omega^2 L} = \frac{1}{4\pi^2 f^2 L} = \frac{1}{4\pi^2\times(60)^2\times 40\times 10^{-3}} = 176\mu F$$

$$V = V_R \Rightarrow X = 110\ V \quad \text{and} \quad I = \frac{V}{R} = \frac{110}{220} = 0.5A$$

Illustration 13

A coil, a capacitor and an *A.C.* source of *rms* voltage 24 V are connected in series. By varying the frequency of the source, a maximum *rms* current 6 A is observed. If this coil is connected to a battery of *emf* 12 V, and internal resistance 4Ω, then calculate the current through the coil.

Solution:

At resonance, current is maximum. $I = \frac{V}{R} \Rightarrow$ Resistance of coil $R = \frac{V}{I} = \frac{24}{6} = 4\Omega$

When coil is connected to battery, suppose I current flow through it then

$$I = \frac{E}{R+r} = \frac{12}{4+4} = 1.5A$$

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Illustration 14

Radio receiver receives a message at 300m band, If the available inductance is 1 mH, then calculate required capacitance

Solution:

Radio receives EM waves (Velocity of EM waves c = 3 x 10^8 m/s)

$\therefore \quad c = f\lambda \Rightarrow f = \frac{3\times10^8}{300} = 10^6\ Hz$

Now $\quad f = \frac{1}{2\pi\sqrt{LC}} = 1\times10^6 \Rightarrow C = \frac{1}{4\pi^2 L\times10^{12}} = 25pF$

Power in AC Circuit

Average power dissipation in LCR AC circuit:

Let $V = V_0 \sin\omega t$ and $I = I_0 \sin(\omega t - \phi)$

Instantaneous power, $P = (V_0 \sin\omega t)\ (I_0 \sin(\omega t - \phi)) = V_0 I_0 \sin\omega t\ (\sin\omega t \cos\phi - \sin\phi \cos\omega t)$

Average power $<P> = \frac{1}{T}\int_0^T (V_0 I_0 \sin^2\omega t\cos\phi - V_0 I_0 \sin\omega t\cos\omega t\sin\phi)dt$

$$= V_0 I_0\left[\frac{1}{T}\int_0^T \sin^2\omega t\cos\phi dt - \frac{1}{T}\int_0^T \sin\omega t\cos\omega t\sin\phi dt\right] = V_0 I_0\left[\frac{1}{2}\cos\phi - 0\times\sin\phi\right]$$

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$\Rightarrow \quad <P> = \frac{V_0 I_0\cos\phi}{2} = V_{rms} I_{rms}\cos\phi$

Instantaneous Average power/actual power/Virtual power/ apparent Peak power

power	dissipated power/power loss	Power/rms	Power
$P = VI$	$P = V_{rms}\, I_{rms}\cos\phi$	$P = V_{rms}\, I_{rms}$	$P = V_0\, I_0$

- $I_{rms}\cos\phi$ is known as active part of current or wattfull current or workfull current. It is in phase with voltage.
- $I_{rms}\sin\phi$ is known as inactive part of current or wattless current or workless current. It is in quadrature (90°) with voltage.

Power Factor:

Average power $\bar{P} = E_{rms} I_{rms}\cos\phi = rms\ \text{power}\times\cos\phi \Rightarrow P_{rms}\times cos\phi$

Power factor ($\cos\phi$) = $\frac{\text{Average power}}{rms\ power}$ and $\cos\phi = \frac{R}{Z}$

Power factor : (i) is leading if I leads V

(ii) is lagging if I lags V

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Golden Key points:

- $P_{av} \le P_{rms}$.
- Power factor varies from 0 to 1
-

Pure/Ideal	ϕ	V	Power factor = cosϕ	Average power
R	0	V, I same Phase	1 (maximum)	$V_{rms}. I_{rms}$
L	$+\frac{\pi}{2}$	V leads I	0	0
C	$-\frac{\pi}{2}$	V lags I	0	0
Choke coil	$+\frac{\pi}{2}$	V leads I	0	0

- At resonance power factor is maximum ($\phi = 0$ so $\cos\phi = 1$) and $P_{av} = V_{rms}\, I_{rms}$

Illustration 15

A voltage of 10 V and frequency 10^3 Hz is applied to $\frac{1}{\pi}$ μF capacitor in series with a resistor of 500Ω. Find the power factor of the circuit and the power dissipated.

Solution:

$\because$ $X_C = \frac{1}{2\pi f C} = \frac{1}{2\pi \times 10^3 \times \frac{10^{-6}}{\pi}} = 500\Omega$

$\therefore$ $Z = \sqrt{R^2 + X_C^2} = \sqrt{(500)^2 + (500)^2} = 500\sqrt{2}\Omega$

Power factor $\cos\phi = \frac{R}{Z} = \frac{500}{500\sqrt{2}} = \frac{1}{\sqrt{2}}$,

Power dissipated = $V_{rms}\, I_{rms}\, \cos\phi = \frac{V_{rms}^2}{Z}\cos\phi = \frac{(10)^2}{500\sqrt{2}} \times \frac{1}{\sqrt{2}} = \frac{1}{10} W$

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Illustration 16

If $V = 100 \sin 100t$ volt and $I = 100 \sin (100t + \frac{\pi}{3})$ mA for an $A.C.$ circuit then find out

(a) phase difference between V and I (b) total impedance, reactance, resistance

(c) power factor and power dissipated (d) components contains by circuits

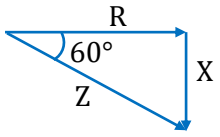
Solution:

(a) Phase difference $\phi = -\frac{\pi}{3}$ (I leads V)

(b) Total impedance $Z = \frac{V_0}{I_0} = \frac{100}{100 \times 10^{-3}} = 1k\Omega$

Now resistance $R = Z\cos 60° = 1000 \times \frac{1}{2} = 500\Omega$

Reactance $X = Z\sin 60° = 1000 \times \frac{\sqrt{3}}{2} = \frac{500}{\sqrt{3}}\Omega$



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(c) $\phi = -60° \Rightarrow$ Power factor = $\cos\phi = \cos(-60°) = 0.5$ (leading)

Power dissipated $P = V_{rms} I_{rms} \cos\phi = \frac{100}{\sqrt{2}} \times \frac{0.1}{\sqrt{2}} \times \frac{1}{2} = 2.5W$

(d) Circuit must contains R as $\phi \neq \frac{\pi}{2}$ and as ϕ is negative

So, C must be there, (L may exist but $X_C > X_L$).

Illustration 17

If power factor of a R-L series circuit is $\frac{1}{2}$ when applied voltage is $V = 100 \sin 100\pi t$ volt and resistance of circuit is 200Ω then calculate the inductance of the circuit.

Solution:

$$\cos\phi = \frac{R}{Z} \Rightarrow \frac{1}{2} = \frac{R}{Z} \Rightarrow Z = 2R \Rightarrow \sqrt{R^2 + X_L^2} = 2R \Rightarrow X_L = \sqrt{3}R$$

$$\omega L = \sqrt{3}R \Rightarrow L = \frac{\sqrt{3}R}{\omega} = \frac{\sqrt{3} \times 200}{100\pi} = \frac{2\sqrt{3}}{\pi} H$$

Illustration 18

A circuit consisting of an inductance and a resistance joined to a 200 volt supply (A.C.). It draws a current of 10 ampere. If the power used in the circuit is 1500 watt. Calculate the wattless current.

Solution:

Apparent power = 200 × 10 = 2000 W

$\therefore$ Power factor $\cos\phi = \frac{\text{True power}}{\text{Apparent power}} = \frac{1500}{2000} = \frac{3}{4}$

Wattless current = $I_{rms} \sin\phi = 10 \sqrt{1 - \left(\frac{3}{4}\right)^2} = \frac{10\sqrt{7}}{4} A$

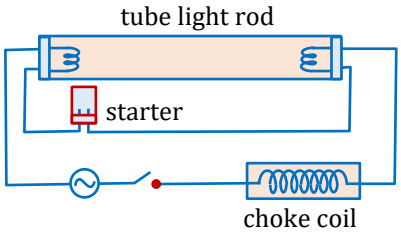
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Choke Coil

In a direct current circuit, current is reduced with the help of a resistance. Hence there is a loss of electrical energy $I^2 R$ per sec in the form of heat in the resistance. But in an AC circuit the current can be reduced by choke coil which involves very small amount of loss of energy. Choke coil is a copper coil wound over a soft iron laminated core. This coil is put in series with the circuit in which current is to be reduced. It also known as ballast.

Circuit with a choke coil is a series L-R circuit. If resistance of choke coil = r (very small)

The current in the circuit $I = \frac{E}{Z}$ with $Z = \sqrt{(R+r)^2 + (\omega L)^2}$ so due to large inductance L of the coil, the current in the circuit is decreased appreciably. However, due to small resistance of the coil r.



The power loss in the choke $P_{av} = V_{rms} I_{rms} \cos\phi \to 0$

$\because$ $\cos\phi = \frac{r}{Z} = \frac{r}{\sqrt{r^2 + \omega^2 L^2}} \approx \frac{r}{\omega L} \to 0$

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Illustration 19

A choke coil and a resistance are connected in series in an *a.c* circuit and a potential of 130 volt is applied to the circuit. If the potential across the resistance is 50 *V*. What would be the potential difference across the choke coil.

Solution:

$$V=\sqrt{V_R^2+V_L^2}\Rightarrow V_L=\sqrt{V^2-V_R^2}=\sqrt{(130)^2-(50)^2}=120V$$

Illustration 20

An electric lamp which runs at 80*V DC* consumes 10 *A* current. The lamp is connected to 100 *V* – 50 *Hz ac* source compute the inductance of the choke required.

Solution:

Resistance of lamp $R=\frac{V}{I}=\frac{80}{10}=8\Omega$

Let *z* be the impedance which would maintain a current of 10 *A* through the Lamp when it is run on 100 Volt *a.c.* then. $Z=\frac{V}{I}=\frac{100}{10}=10\Omega$ but $Z=\sqrt{R^2+(\omega L)^2}$

$\Rightarrow$ $(\omega L)^2 = Z^2 - R^2 = (10)^2 - (8)^2 = 36$

$\Rightarrow$ $\omega L = 6 \Rightarrow L = \frac{6}{\omega}=\frac{6}{2\pi\times 50}=0.02H$

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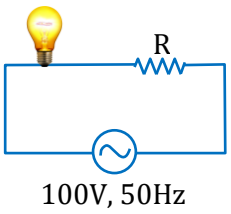
Illustration 21

Calculate the resistance or inductance required to operate a lamp (60*V*, 10*W*) from a source of (100 *V*, 50 *Hz*).

Solution:

(a) Maximum voltage across lamp = 60*V*

$\because$ $V_{Lamp} + V_R = 100$ $\therefore$ $V_R = 40V$

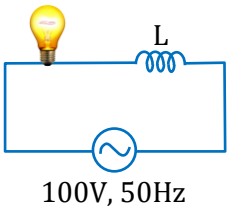


Now current through Lamp is $=\frac{\text{Wattage}}{\text{voltage}}=\frac{10}{60}=\frac{1}{6}A$

But $V_R = IR \Rightarrow 40=\frac{1}{6}(R)\Rightarrow R=240\Omega$

(b) Now in this case $(V_{Lamp})^2 + (V_L)^2 = (V)^2$

$(60)^2 + (V_L)^2 = (100)^2 \Rightarrow V_L = 80\ V$



Also, $V_L = IX_L = \frac{1}{6}X_L$

So, $X_L = 80 \times 6 = 480\ \Omega = L\,(2\pi f) \Rightarrow L = 1.5\ H$

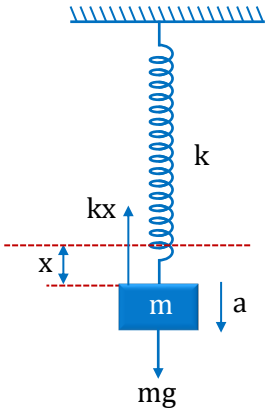
A capacitor of suitable capacitance replace a choke coil in an *AC* circuit, the average power consumed in a capacitor is also zero. Hence, like a choke coil, a capacitor can reduce current in *AC* circuit without power dissipation.

Cost of capacitor is much more than the cost of inductance of same reactance that's why choke coil is used.

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Damped Oscillations

In real life the motion of a simple pendulum, swinging in air, dies out eventually. Why does it happen? This is because the air drag and the friction at the support oppose the motion of the pendulum and dissipate its energy gradually. The pendulum is said to execute **damped oscillations**.



In damped oscillations, the energy of the system is dissipated continuously; but, for small damping, the oscillations remain approximately periodic. The dissipating forces are generally the frictional forces.

To understand the effect of such external forces on the motion of an oscillator, let us consider a system as shown in figure. Here a block of mass m connected to an elastic spring of spring constant k oscillates vertically.

If the block is pushed down a little and released, its angular frequency of oscillation is $\sqrt{\frac{k}{m}}$.

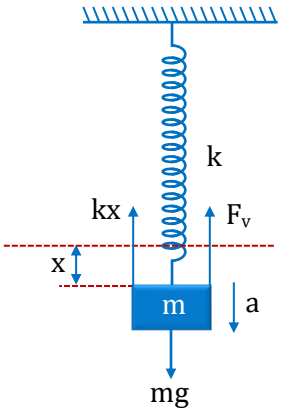
However, in practice, the surrounding medium (air) will exert a damping force on the motion of the block and the mechanical energy of the block-spring system will decrease. The energy loss will appear as heat of the surrounding medium (and the block also).

The damping force depends on the nature of the surrounding medium. If the block is immersed in a liquid, the magnitude of damping will be much greater and the dissipation of energy much faster. The damping force is generally proportional to velocity of the bob and acts opposite to the direction of velocity. If the damping force is denoted by F_d,

$$F_d = -bv$$

Here b is damping coefficient

where the positive constant b depends on characteristics of the medium (viscosity, for example) and the size and shape of the block, etc.



Relation $F_d = -bv$ is usually valid only for small velocity.

When the mass m is attached to the spring and released, the spring will elongate a little and the mass will settle at some height. This position is the equilibrium position of the mass. If the mass is pulled down or pushed up a little, the restoring force on the block due to the spring is $F_S = -kx$, where x is the displacement of the mass from its equilibrium position.

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Thus, the total force acting on the mass at any time t, is

$$F = mg - kx - bv.$$

If a is the acceleration of mass at time t, then by Newton's Law of Motion applied along the direction of motion, we have

$$m\,a = mg - k\,x - b\,v$$

Here we have dropped the vector notation because we are discussing one-dimensional motion.

Using the first and second derivatives of $x(t)$ for $v(t)$ and $a(t)$ respectively, we have

$$m\frac{d^2x}{dt^2} = mg - b\frac{dx}{dt} - k\,x$$

The solution of Eq. (iii) describes the motion of the block under the influence of a damping force which is proportional to velocity. The solution is found to be of the form

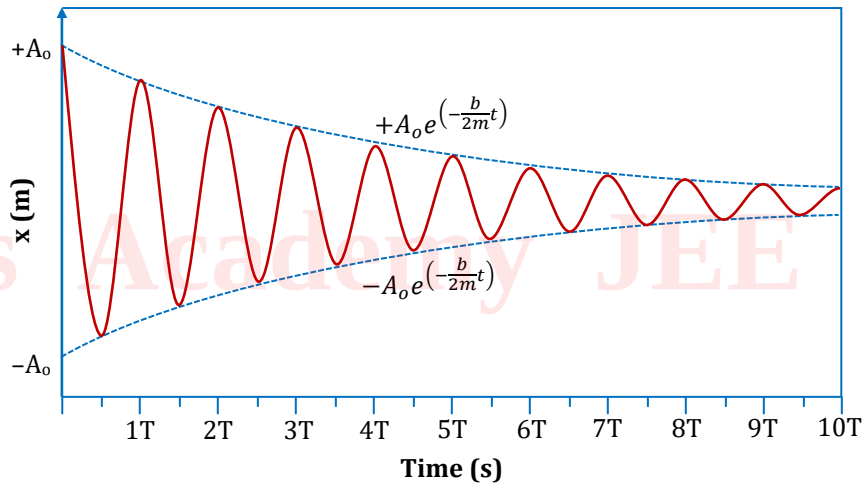
$$x(t) = A_0\, e^{-bt/2m}\, cos\,(\omega t + \phi) \qquad \ldots(i)$$

where A_0 is the initial amplitude and ω is the angular frequency of the damped oscillator given by,

In this function, the cosine function has a period $2\pi/\omega$ but the function $x(t)$ is not strictly periodic because of the factor $e^{-bt/2m}$ which decreases continuously with time.

However, if the decrease is small in one time period T, the motion represented by Eq. (i) is approximately periodic.

The solution, Eq. (i), can be graphically represented as shown in figure.



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Angular Frequency (ω)

$$\omega = \sqrt{\frac{k}{m} - \frac{b^2}{4m^2}}$$

$\omega = \sqrt{\omega_0^2 - \lambda^2}$, where damping constant $\lambda = \dfrac{b}{2m}$

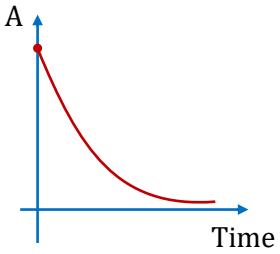
Natural angular frequency $\omega_0 = \sqrt{\dfrac{k}{m}}$

Time Period of Oscillation (T)

$$T = \frac{2\pi}{\omega} = \frac{2\pi}{\sqrt{\omega_0^2 - \lambda^2}} \text{ and natural time period } T_0 = \frac{2\pi}{\omega_0}$$

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Amplitude(*A*)



$$A = A_0 e^{-\frac{b}{2m}t} = A_0 e^{-\lambda t}$$

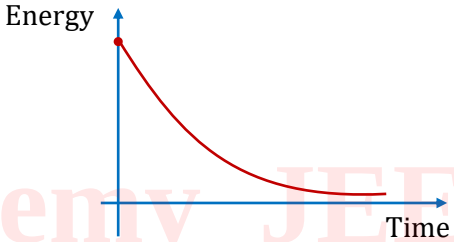
Mechanical Energy

The mechanical energy of the undamped oscillator is $\frac{1}{2}kA^2$.

For a damped oscillator, the amplitude is not constant but depends on time. For small damping, we may use the same expression but regard the amplitude as $A\, e^{-bt/2m}$.

$$E = \frac{1}{2}k\ A^2 = \frac{1}{2}k\ A_0^2 e^{\frac{-b}{m}t}$$

$$E = E_0 e^{-\frac{b}{m}t} = E_0 e^{-2\lambda t}$$



Here E_0 is initial mechanical energy

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$$E_0 = \frac{1}{2}k\ A_0^2$$

(Damping constant for Energy is 2λ)

Above Equation shows that the total energy of the system decreases exponentially with time.

Half Time (t_h)

Half time is time in which amplitude decreases to half of initial Amplitude

$$A = A_0 e^{-\lambda t} = \frac{A_0}{2} \quad \Rightarrow \quad e^{-\lambda t} = 0.5 \quad \Rightarrow \lambda t = \ln(2)$$

$$t_h = \frac{\ln 2}{\lambda} = \frac{0.693}{\lambda}$$

Mean or average Time (t_a)

Average time is time in which amplitude decreases to $\frac{1}{e}$ times of initial Amplitude.

$$A = A_e e^{-\lambda t} = \frac{A_0}{e} \quad \Rightarrow \quad e^{-\lambda t} = e^{-1} \Rightarrow \lambda t = 1$$

$$t_a = \frac{1}{\lambda}$$

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Illustration 22

When an oscillator completes 100 oscillations its amplitude reduced to $\frac{1}{3}$ of initial value. What will be its amplitude, when it completes 200 oscillations:

(A) $\frac{1}{8}$ (B) $\frac{2}{3}$ (C) $\frac{1}{6}$ (D) $\frac{1}{9}$

Solution:

In case of damped vibration, amplitude at any instant is

$a = a_0\ e^{-\lambda t}$

where a_0 = initial amplitude

λ = damping constant

1st Case:

$t = 100\ T$ and $a = \frac{a_0}{3}$

$\therefore \quad \frac{a_0}{3} = a_0\ e^{-\lambda(100T)} \quad \Rightarrow e^{-100\lambda T} = \frac{1}{3}$

2nd Case:

$t = 200\ T$

$a = a_0\ e^{-\lambda t} = a_0\ e^{-\lambda(200T)} = a_0\left(e^{-100\lambda T}\right)^2 = a_0 \times \left(\frac{1}{3}\right)^2 = \frac{a_0}{9}$

Illustration 23

The damping force on an oscillator is directly proportional to the velocity. The units of the constant of proportionality are:

(A) $kgms^{-1}$ (B) $kgms^{-2}$ (C) kgs^{-1} (D) kgs

Ans. (C)

Solution:

$F \propto v \quad \Rightarrow F = kV$

$k = \frac{F}{v} \quad \Rightarrow [k] = \frac{[kgms^{-2}]}{[ms^{-1}]} = kg\ s^{-1}$

Illustration 24

The amplitude of a damped oscillator decreases to 0.9 times its original magnitude in 5*s*. In another 10*s* it will decrease to α times its original magnitude, where α equals.

(A) 0.7 (B) 0.81 (C) 0.729 (D) 0.6

Ans. (C)

Solution:

$A = A_0\ e^{-\frac{b(t)}{2m}}$

After 5 second

$0.9A_0 = A_0 e^{-\frac{b(5)}{2m}}$ ...(i)

After 10 more second

$A = A_0 e^{-\frac{b(15)}{2m}}$ ...(ii)

From (i) and (ii)

$A = 0.729\ A_0$

Hence **Ans. (C)**

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Illustration 25

For the damped oscillator, the mass m of the block is $200g$, $k = 90\ N\ m^{-1}$ and the damping constant b is $40\ gs^{-1}$. Calculate: (a) the period of oscillation, (b) time taken for its amplitude of vibrations to drop to half of its initial value and (c) the time taken for its mechanical energy to drop to half its initial value.

Solution:

(a) The time period T from equation is given by

$$T = \frac{2\pi}{\omega'} = \frac{2\pi}{\sqrt{\frac{k}{m} - \frac{b^2}{4m^2}}} = 0.3\text{sec}$$

(b) Now, from equation, the time, $T_{1/2}$ for the amplitude to drop to half of its initial value is given by

$$A(t) = Ae^{-bt/2m}$$

$$\frac{A}{2} = Ae^{-\frac{bT_{1/2}}{2m}} \quad \Rightarrow \quad T_{1/2} = \frac{2m\ell n2}{b} = 6.93\ s$$

(c) For calculating the time $t_{1/2}$, for its mechanical energy to drop to half of its initial value we make use of equation $E = E_0 e^{-\frac{b}{m}t} = E_0 e^{-2\lambda t}$. From this equation we have

$$E(t_{1/2}) = \frac{E(0)}{2}$$

$$\Rightarrow \quad E(0)e^{-bt/m} = \frac{E(0)}{2}$$

$$t_{1/2} = \frac{(\ell n2)m}{b} = \frac{6.93}{2}$$

This is just half of the decay period for amplitude. This is not surprising, because, according to Equation $E = E_0 e^{-\frac{b}{m}t} = E_0 e^{-2\lambda t}$, energy depends on the square of the amplitude. Notice that there is a factor of 2 in the exponents of the two exponentials.

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Types of Oscillations:

On the basis of the value of λ $\left(\lambda = \frac{b}{2m}\right)$, the damped oscillation can be categorized as

1. Undamped ($\lambda = 0$)

Resistive force is zero, $b = 0$ and $\lambda = 0$

$\Rightarrow$ $A = A_0, E = E_0$ Amplitude and Energy remain constant

$\Rightarrow$ $\omega = \omega_0 = \sqrt{\frac{k}{m}}$ and $T = T_0\sqrt{\frac{m}{k}}$

The system oscillates at its natural angular frequency (ω_0).

2. Under Damped ($\lambda \neq 0$ and $\lambda < \omega_0$)

Resistive force is non-zero, $b \neq 0$ and $\lambda \neq 0$

$\Rightarrow$ Amplitude and Energy decreases

$\Rightarrow$ $A = A_e e^{\lambda t}, E = E_0 e^{-2\lambda t}$

$\Rightarrow$ $\omega = \sqrt{\omega_0^2 - \lambda^2} = +ive$

The system oscillates (at reduced frequency compared to the undamped case) with the amplitude gradually decreasing to zero.

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3. Over Damped ($\lambda \neq 0$ and $\lambda > \omega_0$)

Resistive force is non-zero, $b \neq 0$ and $\lambda \neq 0$

$\Rightarrow$ Amplitude and Energy decreases

$\Rightarrow$ $A = A_0 e^{-\lambda t}, E = E_0 e^{-2\lambda t}$

$\Rightarrow$ As $\omega_0 < \lambda \Rightarrow \omega = \sqrt{\omega_0^2 - \lambda^2}$ is an imaginary number

$\Rightarrow$ Particle finally come to rest without performing any Oscillatory motion

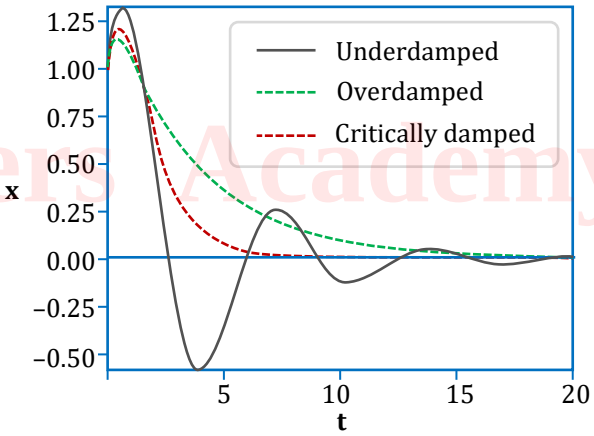
The system returns (exponentially decays) to equilibrium without oscillating. For larger values of the damping constant **λ,** block return to equilibrium more slowly.

4. Critically Damped ($\lambda = \omega_0$)

$\Rightarrow$ Particle finally come to rest without performing any Oscillatory motion

$\Rightarrow$ But It is fastest over damped Oscillation (Particle take smallest time to come to rest)

The system returns to equilibrium as quickly as possible without oscillating.



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Illustration 26

Show that the system $\frac{d^2x}{dt^2} + \frac{dx}{dt} + 3x = 0$ is under damped, find its damped angular frequency

Solution:

$m = 1, b = 1, k = 3$

$b^2 < 4mk$

So, the system is under damped

$$\omega_d = \sqrt{\left(\frac{k}{m} - \frac{b^2}{4m^2}\right)} = \frac{\sqrt{11}}{2}\ s^{-1}$$

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Illustration 27

Determine whether the system is underdamped, overdamped or critically damped.

(i) $\frac{d^2x}{dt^2}+4\frac{dx}{dt}+3x=0$

(ii) $\frac{d^2x}{dt^2}+4\frac{dx}{dt}+4x=0$

Solution:

$$m\frac{d^2x}{dt^2}+b\frac{dx}{dt}+kx=0$$

If $b^2 < 4mk$ then the system is underdamped

If $b^2 > 4mk$ then the system is overdamped

If $b^2 = 4mk$ then the system is critically damped

(i) overdamped

(ii) Critically damped

Power Loss in a Damped Oscillations-Q-factor:

Because of the work done against the nonconservative forces, the energy stored in an oscillator continuously decreases. The fraction of energy lost in a period with respect to its average value for that period is a measure of the quality of the oscillator. The less the value of this fraction is, the more is the ability of the oscillator to sustain periodic motion. Quantitatively, the quality of an oscillator is measured through a Q-factor defined as

$$Q=2\pi\times\frac{\text{Average energy stored in one period}}{\text{average loss of energy in that period}}$$

If the damping is light i.e. $\gamma << \omega_0$, The instantaneous energy of the oscillator is

$$E(t)=\frac{1}{2}mv^2+\frac{1}{2}kx^2=\frac{1}{2}mA^2\omega_0^2e^{-2\gamma t}$$

where we have replaced k by $m\omega_0^2$. At t= 0 the energy of the oscillator is $E_0 = mA^2\,\omega_0^2/2$ so that

$$E(t)=E_0e^{-2\gamma t}$$

$$P(t)=\frac{dE}{dt}=-2\gamma E_0e^{-2\gamma t}=2\gamma E(t)$$

Energy lost in one period can be approximately written as

$$<p(t)>\frac{2\pi}{\Omega}=2\gamma<E(t)>\frac{2\pi}{\Omega}$$

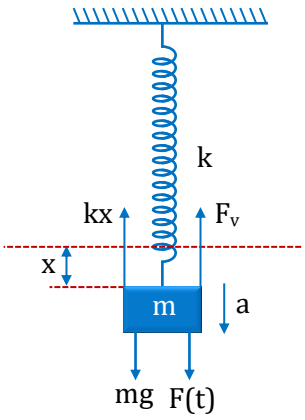
where the angular bracket < ..... > denotes the time average value during a period at time t. The quality or Q-factor for an oscillator is

$$2\pi\frac{<E(t)>}{2\gamma<E(t)>\frac{2\pi}{\Omega}}=\frac{\Omega}{2\pi}\cong\frac{\omega_0}{2\gamma}$$

It may be noted that Q is a dimensionless quantity, greater than unity by our assumption that. $\gamma << \omega_0$

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Forced Oscillations



- An external driving force starts oscillations in a stationary system.
- The amplitude remains constant (or grows) if the energy input per cycle exactly equals (or exceeds) the energy loss from damping.
- Eventually, $E_{driving} = E_{lost}$ and a steady-state condition is reached.
- Oscillations then continue with constant amplitude.
- Oscillations are at the driving frequency ω_d.

$$F(t) = F_0 \cos(\omega_d t + \phi')$$

Oscillating driving force applied to a damped oscillator

Equation for Forced (Driven) Oscillations:

ω_0 = natural frequency $\omega_0 = \sqrt{\frac{k}{m}}$

ω_d = driving frequency of external force

External driving force function:

$$F = F_0 \cos(\omega_d t + \phi')$$

$$F_{net} = F - bv - kx + mg = ma$$

$$F_{net} = F - b\frac{dx}{dt} - kx + mg = m\frac{d^2x}{dt^2}$$

Solution for forced (Driven) Oscillations:

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$$F_{net} = F_0 \cos(\omega_d t + \phi') - b\frac{dx}{dt} - kx + mg = m\frac{d^2x}{dt^2}$$

Solution of this differential equation is

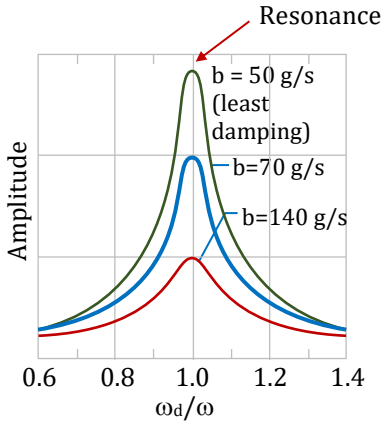
$$X(t) = A\cos(\omega_d t + \phi)$$

Where $A = \frac{F_0/m}{\sqrt{(\omega_d^2 - \omega_0^2)^2 + \left(\frac{b\omega_d}{m}\right)^2}}$

The system always oscillation at the driving always oscillates at the driving frequency ω_d in steady-state.

Resonance Condition in Forced Oscillations

For given values of m, b and F_0 amplitude of motion of block is maximum when $\omega_d = \omega_0$. This is known as Forced Oscillation in Resonance condition



- The largest amplitude oscillations occur at or near resonance ($\omega_D \sim \omega_D$)

  As damping becomes weaker → resonance sharpens and Amplitude at resonance increases.
- At resonance, the applied force is in phase with the velocity and the power $F_0 v$ transferred to the oscillator is a maximum.
- The amplitude of resonant oscillations can become enormous when the damping is weak, storing enormous amounts of energy.

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Applications:

- Buildings driven by earthquakes
- Bridges under wind load
- All kinds of radio devices, microwave
- Other numerous applications

Illustration 28

A particle of mass m is attached to a spring (of spring constant k) and has a natural angular frequency ω_0. An external force $F(t)$ proportional to $\cos \omega t$ ($\omega \neq \omega_0$) is applied to the oscillator. The time displacement of the oscillator will be proportional to:

(A) $\frac{m}{\omega_0^2 - \omega^2}$ (B) $\frac{1}{m\ (\omega_0^2 - \omega^2)}$

(C) $\frac{1}{m\ (\omega_0^2 + \omega^2)}$ (D) $\frac{m}{\omega_0^2 + \omega^2}$

Ans. (B)

Solution:

$$F(t) = F_0 A \cos(\omega t + \phi_0)$$

where $A = \frac{F_0}{\sqrt{m^2\left(\omega_0^2 - \omega^2\right)^2 + \omega^2 b^2}}$

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For small damping

$$A \approx \frac{F_0}{[m(\omega_0^2 - \omega^2)]}$$

$$\Rightarrow \quad x(t) = \left[\frac{F_0}{m(\omega_0^2 - \omega^2)}\right]\cos(\omega + \phi_0)$$

Illustration 29

In forced oscillation of a particle, the amplitude is maximum for a frequency ω_1 of the force, while the energy is maximum for a frequency ω_2 of the force, then:

(A) $\omega_1 = \omega_2$

(B) $\omega_1 > \omega_2$

(C) $\omega_1 < \omega_2$ when damping is small and $\omega_1 > \omega_2$ when damping is large

(D) $\omega_1 < \omega_2$

Ans. (A)

Solution:

Amplitude and energy both are maximum at resonance, when driving frequency is equal to the natural frequency of oscillation. Hence $\omega_1 = \omega_2 = \omega_0$

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Illustration 30

A simple pendulum has a time period T_0 if there is no air resistance. If a small air resistance is acting on the bob as it oscillates,

(A) The time period will be initially more than T_0 and decreases with time.

(B) The time period will be less than T_0 initially and increases with time.

(C) The time period will be less than T_0 and remains constant.

(D) The time period will be more than T_0 and remains constant.

Ans. (D)

Solution:

$$T = \frac{2\pi}{\omega} = \frac{2\pi}{\sqrt{\omega_0^2 - \lambda^2}} \text{ and } T_0 = \frac{2\pi}{\omega_0}$$

$T \geq T_0$ As λ is non-negative

Due to damping Time period of simple pendulum increases in comparison to undamped motion but remains constant with time.

Illustration 31

In damped oscillations, the amplitude after 10 sec. is 0.6 A_0, where A_0 is the initial amplitude, then determines the amplitude after another 10 sec.

Solution:

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$$A = A_0 e^{-\lambda t}$$

At $t = 10$ $A = A_0 e^{-10\lambda} = 0.6A_0$

$\Rightarrow$ $e^{-10\lambda} = 0.6$

At $t = 20$; $A = A_0 e^{-20\lambda}$

At $t = 20$; $A = A_0 (0.6)^2 = 0.36A_0$

Illustration 32

In damped oscillations, the amplitude after 10 sec. 0.6 A_0, where A_0 is the initial amplitude, then determine the value of damping constant (in s^{-1}). Given ln(5) = 1.61, ln (3) = 1.1.

Solution:

$$A = A_0 e^{-\lambda t}$$

At $t = 10$; $A = A_0 e^{-10\lambda} = 0.6A_0$

$\Rightarrow$ $e^{-10\lambda} = 0.6$

$\Rightarrow$ $-10\lambda = \ln(0.6) = \ln\frac{3}{5} = \ln 3 - \ln 5$

$\Rightarrow$ $-10\lambda = 1.1 - 1.61 = -0.5 - 10\lambda$

$\Rightarrow$ $\lambda = 0.05 s^{-1}$

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Illustration 33

Damped harmonic oscillator consists of a block (m= 2 kg), a spring $\left(k=50\pi^2 N/m\right)$ and a damping force $\left(F=-12\pi v N\right)$. Find angular frequency and time period of motion. Also find angular frequency and time period if damping force is removed.

Solution:

$$\omega=\sqrt{\frac{k}{m}-\left(\frac{b}{2m}\right)^2}=\sqrt{\omega_0^2-\lambda^2}$$

m = 2 Kg, k = $50\pi^2$ N/m and b = 12 π

$$\omega=\sqrt{\frac{50\pi^2}{2}-\left(\frac{12\pi}{4}\right)^2}=\sqrt{25\pi^2-9\pi^2}=12\pi\, Rad/s$$

$$T=\frac{2\pi}{\omega}=\frac{2\pi}{12\pi}=\frac{1}{6}\text{sec.}$$

For undamped motion,

$$\omega_0=\sqrt{\frac{k}{m}}=\sqrt{\frac{50\pi^2}{2}}=5\pi\, Rad/s$$

$$T_0=\frac{2\pi}{\omega}=\frac{2\pi}{5\pi}=\frac{2}{5}\text{sec}$$

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Illustration 34

If the differential equation given by the oscillatory motion of body in a dissipative medium under the influence of a periodic force, then the state of maximum amplitude of the oscillation is a measure of

(A) Free vibration (B) Damped vibration (C) Forced vibration (D) Resonance

Ans. (D)

Solution:

In Resonance condition amplitude of forced oscillation is maximum.

Illustration 35

When an automobile move over an uneven path, it experience jerks. To minimize the effect of the jerks the suspensions used in should satisfy the condition of

(A) Undamped Oscillation (B) Over-damped Oscillation

(C) Under-damped Oscillation (D) Critical damped Oscillation

Ans. (D)

Solution:

When an automobile move over an uneven path, it experience jerks. To minimize the effect of the jerks the suspensions used in should satisfy the condition of Critical damped Oscillation, because Automobile require to come its natural state as soon as possible.

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